MARKING SCHEME
BIOLOGY ZONAL EXAM (ARUSHA & MANYARA)

SECTION 'A': 16 marks

1.

Question	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answers	A	A	B	C	C	C	B	C	D	B

01@10 = 10marks

2.

COLUMN 'A'	i	ii	iii	iv	v	vi
COLUMN 'B'	D	A	G	E	F	H

01@6 = 06marks

SECTION B (54marks)

3 (a)(i)

Roman	Field	branch of biology
(i)	Medicine	Anatomy
(ii)	Agriculture	parasitology
(iii)	forestry	Ecology

Field – 03mark
branch – 01 mark
=06marks

(b) 3 components of coordination and their examples

01. RECEPTORS eg All sense organs (i.e eyes, ears, skin, tongue & noses)
02. COORDINATORS eg central nervous system i.e Brain & Spinal cord.
03. EFFECTORS eg Glands and muscles.

component – 01/2 mark
example – 01/2 mark
⇒ 03

4(a)(i) sweep net

(ii) Scoop net

01 Mark @4 = 4 Marks

(iii) Quadrat.

(iv) cage

(b) The question demand the disadvantages of artificial classification system, which include:

(i) Closely related organisms are placed into different group eg human being and bat.

(ii) Unrelated organisms may be placed in the same groups eg birds and bats or insects and bats.

(iii) It is less accurate, because it use only few observabl characteristics

(iv) It create confusion among biologists from differen parts of the world.

(v) It provide us with limited information about the organisms.

(vi) It does not allow the prediction of information hence limit more advancement in taxonomy.

Any 5 points @ 1marks = 05 m

5(a) (i) To reduce loss of water from the plant living in deserts.

(ii) To ensure maximum absorption of the digested food materials.

(iii) For trapping dust and germs entering the tracheal system.

(iv) To overcome/reduce air resistance

(v). so that the gases diffuse into the cell insolution form.

(b) (i) Hinge joint

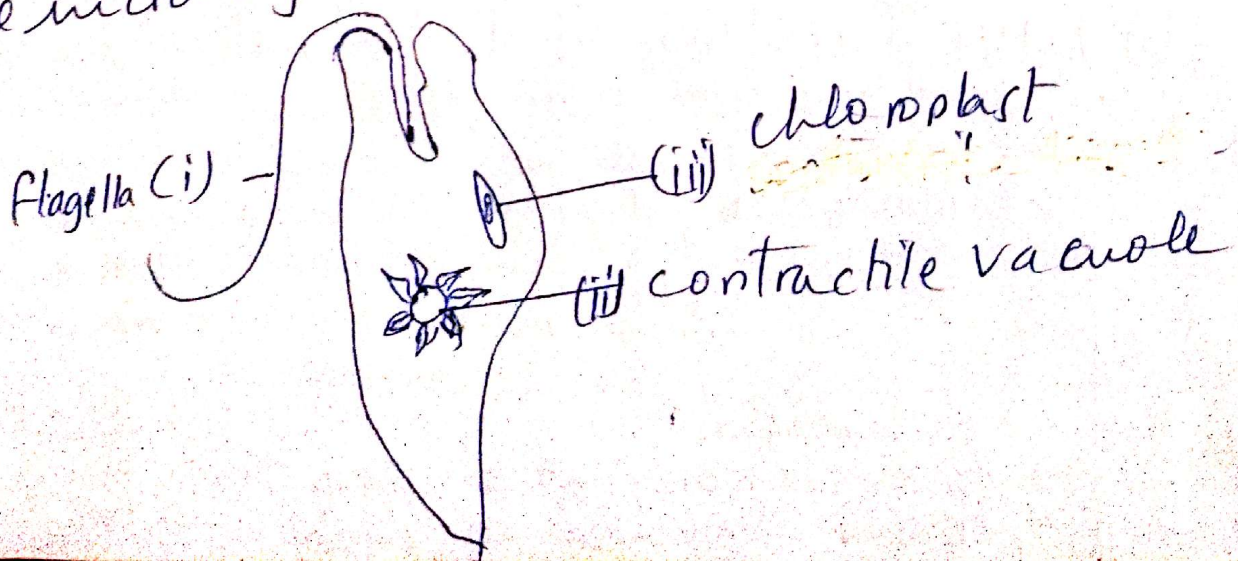
(ii) ~~ball and socket joint~~ Hinge joint

(iii) Ball and socket joint

(iv) Ball and socket joint.

6(a) The micro-organisms were Euglena sp.

(b)



(b) Three types of meristems

(i) APICAL MERISTEMS – occurs at the tips of the shoot and roots. It cause growth in length of roots and stems. ∴ primary growth.

(ii) LATERAL MERISTEMS: – occurs in the – Vascular cambium – and cause growth in girth. ∴ secondary growth

(iii) INTERCALARY MERISTEMS – These occurs in the region of permanent tissues eg

7a

Food item	Food nutrient found	
• Egg	Protein	01 mark
• Irrish potatoes	starch	01 mark
• Sunflower seed	Lipid	01 mark

(b) Four functions of the food nutrient found in egg

Food nutrient found in egg is protein

Function of protein

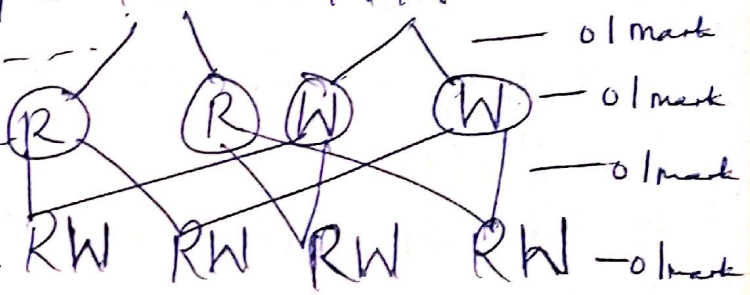
→ Building body cells
→ promoting growth
→ making protoplasm
→ To repair damaged tissues
→ Transportation of other molecules across the cell membrane
→ synthesis of enzymes + haemoglobin
→ synthesis of hormones, and antibodies
→ Source of energy during starvation
→ Aid in contraction of muscles

any 4 points @ $1\frac{1}{2}$ = 06

8. Let the gene for red flowered rose plant — 0l mark
be 'R' and gene for white flowered rose plant
be 'W'

Parental phenotype: Red x White — 0l mark

Parental genotype RR x WW — 0l mark



∴ All F1 generations are RW genotypically and Pink Phenotypically. — 0l mark

0l mark

Contributions in Industries. The knowledge of genetics has been widely used in Industries to manufacture different products.
Living organisms especially microorganisms like bacteria and yeast are used in — production of p breweries, beverages and drugs.

Application of genetics in genetic counsellin g. Genetic counseling is the process of advising and guiding individuals or families affected by or at risk of genetic disorders and — helping them to adapt to the situation.

Application in protection and conservation of genetic resources.
Genetic resources are genetic materials — actual or potential values that safeguard the survival of a particular species and global biodiversity
→ Application in blood transfusion
→ Contribution in establishing — parternety.

Introd — 02 marks
;; Any 6 points @ 02 marks
point — 01, explanation — 01 mark
conclusion → 01 mark.

10. Introd.

Temperature regulation is the ability of the organisms to maintain relatively constant body temperature regardless of the changes of external temperature. Organisms can regulate temperature in their bodies by two ways, namely behavioral means and physiological means.

Here below are the behavioral means (mechanisms) of temperature regulation in poikilotherms.

Poikilotherms are organisms whose their body temperature changes with the changes in environmental temperature

Temperature regulation in poikilotherms is mostly done by behavioral means which include:

Sun basking is the common means of warming up body using sun rays

They do this by exposing their bodies on sun so that they absorb heat from the sun rays by radiation

Hibernation: Is a state where animals undergo ... Long period of deep sleep in winter. During this period, they reduces their metabolic activities and slows the heart beat rate for purpose of conserving energy.

Migration: Animals with no suitable insulators like fur and fat cannot hibernate. Instead they have to migrate to the warmer places

Aestivation: Is the form of hibernation during hot condition where by animals go into the state of dormancy to reduce body metabolic activities. During hot condition, some animals hide in nest or burrows.

Burrowing: During hot condition the animals tend to hide in burrows or shady or cool places to protect themselves from high temperature.

Huddling: Is the mutualistic gathering or crowding together of animals in a close mass. This body contact increases body temperature.

Conclusion: 01 mark

Introd - 02 marks
point - 01 mark
Explanation - 01 mark.
Any 6 points = 12 marks.

11. Introduction
Meaning of irresponsible sexual behaviour.
These are those sexual practices that are not legally, socially and morally accepted by the society. They include: practicing unsafe sex; having many sexual partners; unfaithfulness to one's sexual partner, unaccepted exposure of private parts in public; prostitution and rape & incest

Here are the methods of reducing irresponsible sexual behaviours in order to remain safe.

- Choose good friends carefully
- Avoid peer pressure.
- Adopt good practices and attitudes from people who are well respected.
- Observe and obey school rules and regulations.
- Be assertive and make informed decisions.
- Having appropriate knowledge and facts about reproductive health
- self control and self ..... discipline
- Imposing strict government laws against rape and sexual abuse cases

Conclusion - 01 Mark

Introd - 02 marks

Mainbody - point - 01 mark
Expl - 01 mark

any 6 points = 12 marks